

VAMSHIKRISHNA KATAKAM

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AI and Software Engineering graduate with hands-on experience in Machine Learning, Computer Vision, Generative AI, and Accessibility Systems. Experienced in developing intelligent multimodal applications involving speech processing, computer vision, UI automation, and AI-driven automation. Passionate about building impactful AI solutions that solve real-world accessibility challenges.

EDUCATION

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD, TELANGANA	2022 – 2026
<i>B.E in Artificial Intelligence and Data Science, GPA: 8.92/10</i>	
<i>Sri Chaitanya Junior College, Intermediate(Class XII) 96.7%</i>	<i>2020 – 2022</i>
<i>Jayarani High School, SSC(Class X) GPA 10/10</i>	<i>2020</i>

SKILLS AND OTHERS

Programming Languages: Python, SQL

AI/ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI, Large Language Models (LLMs)

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, MediaPipe, Pandas, NumPy, Matplotlib

Tools & Technologies: Git, GitHub, VS Code, Windows UI Automation, MySQL

Core Subjects: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

Languages Known: English, Telugu, Hindi

PROJECTS

Multimodal AI Accessibility System May 2025 – Present

Python, PyTorch, OpenCV, Whisper, Windows UI Automation, Computer Vision

- Developing a multimodal AI accessibility system for visually impaired and motor-disabled users.
- Implemented voice-controlled desktop interaction using speech recognition and natural language processing.
- Built intelligent screen reading, UI automation, and semantic navigation modules.
- Integrated computer vision, OCR, and object detection for enhanced accessibility features.

AI-Based Resume Shortlisting System March 2025

Python, NLP, Scikit-learn

- Automated resume screening using NLP techniques to extract skills, experience, and relevant keywords.
- Applied machine learning models in Scikit-learn for candidate ranking and shortlisting.
- Reduced manual review effort, improving recruitment speed and accuracy.

Hand Gesture-Based Image Control System June 2025

OpenCV, MediaPipe

- Built a real-time image control system using OpenCV and MediaPipe.
- Enabled zoom, pan, rotate, and image switching through hand gestures.
- Optimized the interface by eliminating black borders and restricting actions within image bounds.

RESEARCH PUBLICATION

"AI-Powered Personal Assistant for Differently Abled Individuals"

Accepted at the 2nd International Conference on Emerging Technologies in Engineering and Pharmacy Life Sciences (ICETEPLS-2026)

- Proposed an AI-powered assistive system to enhance accessibility and independence for differently abled individuals.

CERTIFICATIONS & ACHIEVEMENTS

- Certified as an AI Associate by Salesforce and completed NPTEL courses in Data Analytics with Python and Fundamentals of Object-Oriented Programming.
- Solved 150+ problems on LeetCode and actively participated in multiple hackathons and coding competitions.